

ELGIN 5 S, AZ, USA

WMO: 722760

Lat: 31.5907N

Lon: 110.5087W

Elev: 4811

StdP: 12.31

Time Zone: -7.00 (W07)

Period: 02-19

WBAN: 53132

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	20.6	24.0	-3.3	5.5	64.3	0.8	6.8	59.3	17.8	49.7	15.2	48.8	2.0	N/A	0.411

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	32.0	95.2	58.7	92.9	59.2	90.6	59.2	67.5	80.7	66.5	79.4	65.9	78.6	7.8	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.3	108.1	70.2	63.3	104.4	69.8	62.3	100.7	69.7	34.8	80.6	34.1	79.5	33.4	78.2	71.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.9	13.0	11.5	DB	13.1	98.9	4.3	2.2	10.0	100.4	7.5	101.7	5.1	103.0	2.0	104.6	(4)
(5)			WB	10.0	69.1	4.8	1.0	6.6	69.8	3.8	70.4	1.1	70.9	-2.4	71.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.2	44.8	46.9	52.7	57.9	65.2	75.3	76.0	73.7	69.9	61.7	52.4	44.8		
	DBStd	12.66	6.16	7.02	6.52	5.97	5.94	4.02	3.60	3.05	3.58	5.50	6.52	6.67		
	HDD50	603	176	127	48	12	1	0	0	0	0	2	50	187		
	HDD65	2942	625	507	380	220	72	0	0	0	6	126	379	627		
	CDD50	4317	16	40	133	247	471	760	807	733	597	367	121	25		
	CDD65	1180	0	0	0	5	76	310	342	268	154	25	0	0		
	CDH74	15416	0	26	140	453	1787	4687	3313	2401	1796	728	83	2		
	CDH80	6483	0	1	9	67	633	2524	1551	938	615	141	4	0		
(14) Wind	WSAvg	4.8	4.4	5.0	5.5	6.1	6.1	5.6	4.6	3.7	3.8	4.3	4.4	4.4		
(15) Precipitation	PrecAvg	17.3	1.3	1.1	0.9	0.4	0.2	0.6	4.2	3.9	1.6	0.7	0.8	1.3		
	PrecMax	23.9	7.0	3.8	3.6	1.8	1.3	2.6	8.0	6.3	3.7	7.4	3.3	4.1		
	PrecMin	10.2	0.0	0.0	0.0	0.0	0.0	0.0	1.3	1.8	0.6	0.0	0.0	0.0		
	PrecStd	4.1	1.7	0.9	0.9	0.6	0.3	0.6	1.8	1.1	0.7	1.5	0.7	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.3	77.6	81.2	85.1	92.3	99.0	97.0	93.7	91.3	87.0	80.0	73.2		
		MCWB	46.1	47.8	49.8	52.1	54.9	58.6	61.1	64.1	61.1	55.8	50.0	47.0		
	2%	DB	67.9	73.2	77.9	81.9	89.0	96.2	94.3	90.5	88.6	83.7	76.5	69.0		
		MCWB	45.1	46.4	48.6	50.9	53.8	57.9	61.8	63.9	60.8	55.3	50.2	45.9		
	5%	DB	64.8	68.9	74.9	79.4	86.5	94.1	91.5	88.2	86.3	81.2	73.3	64.9		
		MCWB	43.9	45.4	47.6	49.9	52.7	57.4	62.2	64.0	60.9	54.9	49.5	45.5		
	10%	DB	61.1	64.3	71.6	76.4	83.8	91.8	88.8	85.8	84.0	78.5	69.8	60.5		
		MCWB	42.4	43.9	46.3	48.8	51.7	57.2	62.6	63.8	60.6	54.1	48.2	43.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.8	51.2	51.4	54.3	56.7	65.0	68.4	69.2	67.1	62.2	55.4	51.7		
		MCDB	64.5	61.0	75.7	80.6	86.9	80.7	81.3	82.9	78.6	75.4	69.0	59.3		
	2%	WB	47.9	49.5	50.0	52.5	55.0	63.2	67.3	67.9	65.8	60.1	52.6	49.1		
		MCDB	61.6	64.1	74.2	78.9	85.5	82.5	79.7	81.8	77.7	75.8	67.3	63.1		
	5%	WB	46.0	47.7	48.7	51.2	53.4	62.0	66.4	67.0	64.8	58.5	51.1	47.3		
		MCDB	60.9	65.2	72.1	76.9	82.9	82.3	78.8	80.6	76.9	74.5	67.6	62.3		
	10%	WB	44.1	45.8	47.5	49.9	52.0	60.5	65.7	66.2	63.8	56.4	49.7	45.4		
		MCDB	59.1	63.6	69.6	75.0	80.4	83.0	78.1	79.4	76.2	72.4	66.8	59.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	29.2	29.8	31.7	32.4	32.8	32.0	23.6	23.4	26.7	30.8	30.9	28.2		
		MCDBR	35.2	38.0	37.5	37.4	36.1	34.9	28.4	27.8	31.4	34.2	35.6	34.4		
		MCWBR	17.7	17.7	16.7	16.3	15.0	12.8	7.8	8.2	10.7	14.0	16.1	16.8		
	5% WB	MCDBR	30.0	31.9	35.1	33.9	34.6	28.4	21.7	22.5	23.3	29.4	29.4	28.7		
		MCWBR	15.6	15.2	15.3	15.0	13.9	9.8	6.4	7.5	8.6	12.3	14.2	15.2		
(40) Clear-Sky Solar Irradiance	taub		0.244	0.247	0.264	0.272	0.286	0.311	0.363	0.351	0.313	0.279	0.254	0.243		
	taud		2.683	2.676	2.597	2.558	2.555	2.487	2.463	2.488	2.572	2.620	2.653	2.675		
	Ebn at Noon		321	327	326	324	317	307	291	294	304	311	314	316		
	Edh at Noon		23	25	30	32	32	35	36	34	30	26	23	22		
(44) All-Sky Solar Radiation	RadAvg		1177	1474	1945	2378	2605	2540	2035	1978	1846	1619	1297	1081		
	RadStd		79	119	108	82	102	151	148	79	93	102	63	49		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-135	N/A	N/A

Nomenclature: See separate page